

Constraint-threshold conjunction: Real Stability

Research Architecture Runtime
operator/constraint-threshold-conjunction

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Abstract

`constraint-threshold-conjunction` preserves a Boolean combination of threshold constraints under buffered scalar noise. Lean `LEAN_FULL`.

This theorem is: Reduction.

1 Problem

Conjunction or disjunction of threshold predicates on coordinates/scores.

2 Stability notion

Each literal is threshold-stable; Boolean combinations inherit preservation.

3 Definitions

Reduce to multi-threshold / threshold preservation on each cut.

4 Theorem

Theorem 1. *If every atomic threshold is buffer-stable for (x, x', ε) , the conjunction/disjunction value is preserved.*

5 Intuition

Boolean operations are deterministic functions of stable bits.

6 Examples

Example 2. *Two cuts $x \geq 1$ and $x \geq 3$ with $x = 4$, $\varepsilon = 0.2$: both bits stable, so their conjunction stays true.*

7 Proof

Each atomic predicate is preserved by scalar threshold preservation under the buffer hypotheses. Conjunction (resp. disjunction) is a fixed Boolean function of those bits, hence preserved (Theorem 1).

8 Formal statement

Lean module: `Research.Operators.ConstraintThresholdConjunction.Preservation`. Source file: `lean/Research/Operators/ConstraintThresholdConjunction/Preservation.lean`. Theorems: `constraint_threshold_conjunction_conjunction_preservation`, `constraint_threshold_conjunction`. Certificate: `lean/certificates/constraint-threshold-conjunction/constraint-threshold-conjunction-`. Status: `LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

9 Proof dependencies

- Threshold / multi-threshold cores.
- Boolean congruence.

10 Consequences

Constraint-side companion to multi-threshold counts.